(a) Progressive pairwise evidence



RGB



Infrared

Example input: S02 / 000009

MINIMA / RoMa correspondences + robust fitting

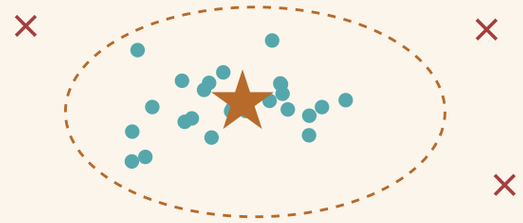
Frame homography H_i
Geometry-informed weight q_i
r_i, c_i, e_i, m_i, n_i

Evenly spaced evidence; progressive expansion
All pairs for $N \leq 300$; staged expansion otherwise

(b) Robust scene consensus

Median / MAD filtering

Accepted H_i, q_i



Parameter-space schematic

Quality-weighted aggregation

$\bar{h}_S = \sum_{i \in C_S} q_i h_i \,/\, \sum_{i \in C_S} q_i$

Scene $\bar{H}_S$

● Kept hypothesis × Rejected ★ Consensus

(c) Cross-modal verification and operating profile

Scene $\bar{H}_S$

Representative QA probes
Edge / gradient improvement
Relative outliers + frame support
Geometry + consensus diagnostics

Fixed canonical gate
Support AND (L_S OR W_S)
Retained / QA-filtered product

Reliability score R(S)
Diagnostic ranking
Reliability-coverage operating profile

Canonical selection and score ranking are separate outputs.